ABOUT ME

C/C++ Software Engineer with 4+ years of experience in Linux and QNX-based systems, system-level debugging, software integration, and asynchronous task processing. Strong understanding of C++ memory management, multithreading, synchronization, algorithms, and data structures. Experienced in troubleshooting complex runtime issues and developing reliable software and build infrastructure.

SKILLS

PROGRAMMING: C++, C, JAVA, PYTHON, SHELL SCRIPTING

C++17/20: STL, OBJECT-ORIENTED PROGRAMMING, TEMPLATES, SMART POINTERS, MEMORY MANAGEMENT, MULTITHREADING, SYNCHRONIZATION

ALGORITHMS & DATA STRUCTURES: HASH TABLES, TREES, GRAPHS, HEAPS, BFS/DFS, DIJKSTRA, COMPLEXITY ANALYSIS

SYSTEMS PROGRAMMING: LINUX, QNX, CONCURRENCY, ASYNCHRONOUS TASK PROCESSING

DEBUGGING & ANALYSIS: GDB, TRACE32, CANOE, SYSTEM LEVEL DEBUGGING, XCP

BUILD SYSTEMS: MAKE, CMAKE, BAZEL AND CONAN; DEPENDENCY MANAGEMENT AND RELEASE BUILDS

CI/CD & TOOLING: GITHUB ACTIONS, JENKINS, DOCKER, QEMU AND GIT

LINKS

GitHub:
https://github.com/donKljestan

LANGUAGES

SERBIAN

ENGLISH

NIKOLA KLJESTAN

C/C++ SYSTEMS SOFTWARE ENGINEER

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WORK EXPERIENCE

SYNCHROTEK
Feb 2026 - Present

C/C++ Embedded Software Engineer

Maintain and enhance production C/C++ software for automotive embedded systems across AUTOSAR Classic and Adaptive platforms.
Analyzed and optimized C++ runtime performance and memory usage, identifying bottlenecks and implementing targeted improvements.
Migrate software components from CMake-based builds to a Bazel monorepo, defining build targets, dependencies, and integration rules.
Further develop and maintain a C++ asynchronous task management component, applying multithreading, synchronization, and memory-management concepts.
Review C/C++ code and pull requests as a code owner, ensuring code quality and integration readiness.

NXP
Feb 2025 - Jan 2026

Software Engineer

Developed and maintained Linux-based build and release infrastructure using Make, CMake, Conan, and Bazel.
Created and maintained build configurations, targets, and component dependency definitions using Make and CMake.
Automated build and integration workflows using Python and GitHub Actions, improving validation and reducing manual effort.
Developed and executed Linux-based build, test, and simulation workflows using Docker, QEMU, shell scripting, and Python.
Adapted existing C++ components to changes in platform APIs, dependencies, and build environments.

TTTECH AUTO
Dec 2021 - Jan 2025

Software Engineer

Debugged embedded C/C++ software on target hardware using TRACE32, CANoe, and system-level validation tools.
Performed software/hardware integration across ECU platforms, including QNX-based systems and inter-component communication.
Modified component interfaces, configuration handling, and runtime behavior to support integration on new ECU targets.
Diagnosed and fixed C++ compilation, linking, and runtime issues on QNX-based embedded targets.
Resolved a cross-component communication issue by tracing data flow through multiple C++ modules and implementing a targeted fix.
Collaborated with distributed software, system, and hardware teams across different locations to coordinate changes and resolve integration issues.

EDUCATION

FACULTY OF ELECTRICAL ENGINEERING
2022

Bachelor of Science

Major: Computer Science and Informatics - Computer Engineering
Graduation thesis: *Implementation of a system for analyzing and processing data collected from parking sensors*